Distributed Systems Staff Engineering Cheatsheet

Essential Formulas, CAP Tradeoffs, Latency Numbers & Consistency Invariants

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1. Latency Numbers Every Staff Engineer Must Know

L1 Cache Reference: 0.5 ns

Mutex Lock / Unlock: 25 ns

Main Memory [RAM] Read: 100 ns

NVMe SSD Random Read: 10,000 ns [10 us]

Send 2KB over 1 Gbps Commodity Network: 20,000 ns [20 us]

Round-trip within Same Datacenter: 500,000 ns [0.5 ms]

Cross-Continent Ping [SF to NYC]: 40,000,000 ns [40 ms]

Transpacific Ping [SF to Tokyo]: 120,000,000 ns [120 ms]

2. Consensus & Quorum Equations

Strict Quorum Rule: R + W > N [Guarantees Read-Your-Writes Consistency]

Raft Leader Election Majority: Quorum = floor[N / 2] + 1

Fault Tolerance: An N-node cluster tolerates at most F = floor[[N - 1] / 2] node failures

PACELC Principle: If Partition [P], choose Availability [A] or Consistency [C]; Else [E], choose Latency [L] or Consistency [C]

3. Distributed Transactions vs Sagas

Two-Phase Commit [2PC]: High consistency, blocking, single coordinator failure halts cluster

Saga Pattern: Choreographed or Orchestrated event sequence with compensating transactions